Audio Feature Analysis Report

Parkinson's Disease vs. Concussion (TBI) Classification

Date: April 21, 2026

Dataset: 114 PD samples · 131 TBI samples · 378 audio features

Models: Logistic Regression · Decision Tree · Random Forest · LDA

Validation: 10-fold Stratified Cross-Validation

Metric	Value
PD samples	114
TBI samples	131
Total features	378
Features $p < 0.01$	226
Features $p < 0.005$	212
Best accuracy (CV)	100.0%
Best ROC-AUC (CV)	100.0%

1. Overview

This report presents a comprehensive analysis of audio features extracted from speech recordings to distinguish between **Parkinson's Disease (PD)** and **Traumatic Brain Injury / Concussion (TBI)**. The pipeline covers three stages: (1) statistical significance testing, (2) dimensionality-reduction visualisation, and (3) a supervised machine-learning classification pipeline with robust cross-validation.

Both CSV datasets were loaded and combined into a single design matrix. Metadata columns (file_name, audio_path, patient_id, date, time) were excluded, leaving **378 numeric audio features** per sample.

2. Statistical Analysis

For each of the 378 features, a **Shapiro-Wilk normality test** was first applied to both groups. Features where both groups passed normality ($p > 0.05$) were tested with Welch's two-sample **t-test**; all others were tested with the non-parametric **Mann-Whitney U test**. Results were ranked by ascending p-value.

2.1 P-value Distribution

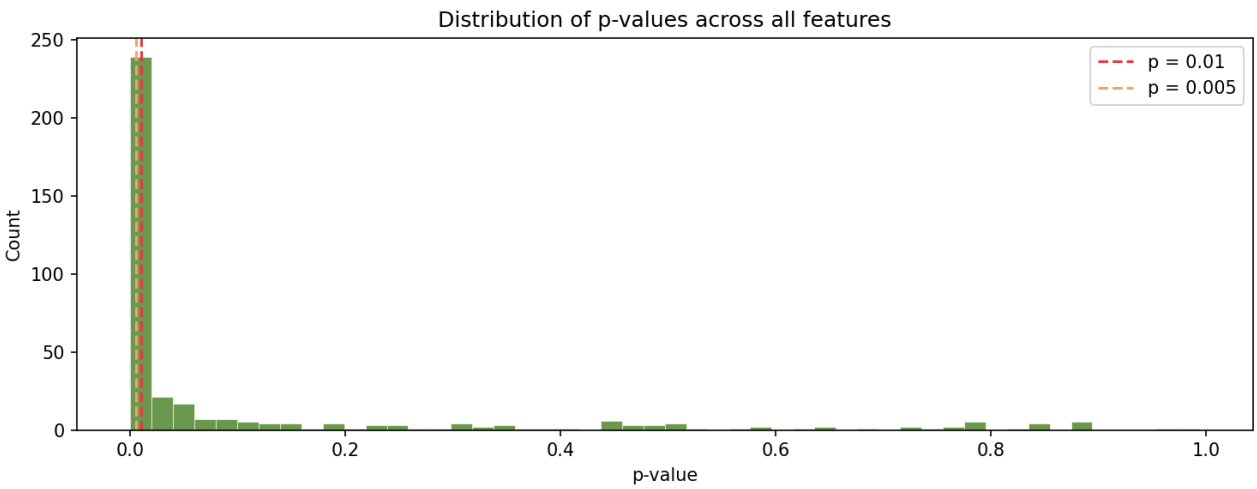

Figure 1. Distribution of p-values across all 378 features. Dashed red line = p 0.01; dashed orange line = p 0.005.

2.2 Significance Summary

Threshold	# Significant Features	% of Total
$p < 0.01$	226	59.8%
$p < 0.005$	212	56.1%

2.3 Top-20 Most Significant Features

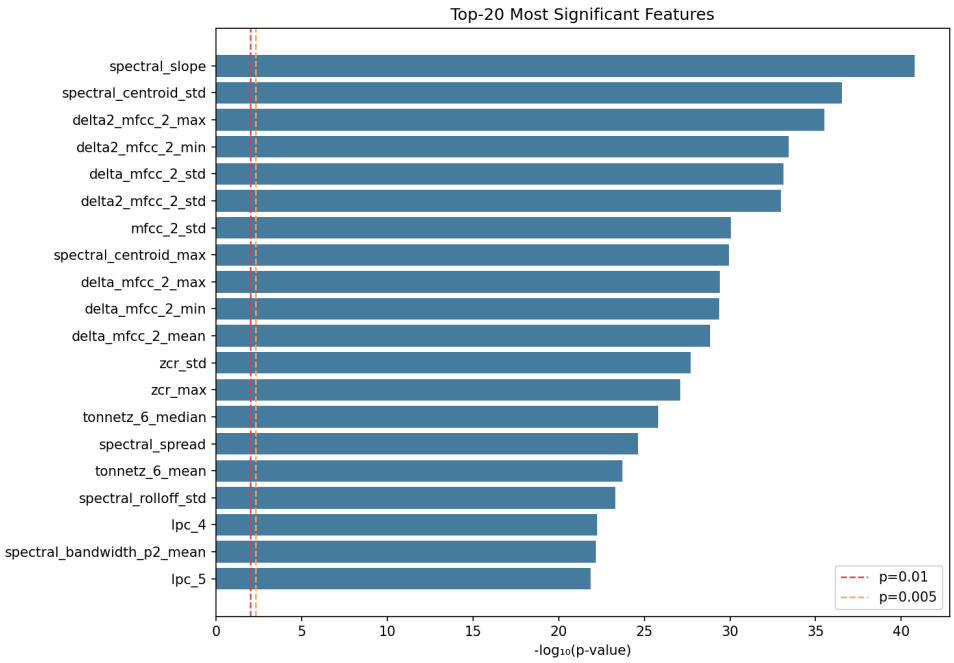

Figure 2. Top-20 features ranked by –log■■(p-value).

2.4 Top-25 Significant Features Table (p < 0.005)

Feature	Test Used	p-value	PD Mean	TBI Mean
spectral_slope	Mann-Whitney U	1.69e-41	-0.0019	-0.0009
spectral_centroid_std	t-test	2.82e-37	454.2046	686.7518
delta2_mfcc_2_max	t-test	3.01e-36	13.1366	18.1590
delta2_mfcc_2_min	t-test	3.69e-34	-11.4678	-15.5434
delta_mfcc_2_std	Mann-Whitney U	7.34e-34	5.9158	8.6918
delta2_mfcc_2_std	Mann-Whitney U	1.07e-33	3.4965	5.1515
mfcc_2_std	Mann-Whitney U	8.58e-31	32.3030	44.7868
spectral_centroid_max	t-test	1.10e-30	3150.9980	3898.7098
delta_mfcc_2_max	t-test	3.78e-30	20.2334	27.0051
delta_mfcc_2_min	t-test	4.14e-30	-19.3754	-26.1619
delta_mfcc_2_mean	Mann-Whitney U	1.39e-29	0.0138	-0.0368
zcr_std	Mann-Whitney U	2.00e-28	0.0482	0.0826
zcr_max	t-test	7.81e-28	0.3560	0.4960
tonnetz_6_median	Mann-Whitney U	1.53e-26	-0.0154	0.0144
spectral_spread	Mann-Whitney U	2.38e-25	1192.0639	1402.6363
tonnetz_6_mean	Mann-Whitney U	1.85e-24	-0.0118	0.0155
spectral_rolloff_std	t-test	4.73e-24	974.7929	1194.6882
lpc_4	Mann-Whitney U	5.89e-23	2862.9674	-3177.3999
spectral_bandwidth_p2_mean	Mann-Whitney U	6.92e-23	1157.5612	1266.1764

Feature	Test Used	p-value	PD Mean	TBI Mean
lpc_5	Mann-Whitney U	1.34e-22	-4292.6793	3547.6980
mfcc_2_min	Mann-Whitney U	1.52e-22	22.9939	-12.7247
lpc_6	Mann-Whitney U	4.91e-22	4363.5145	-3140.4275
lpc_3	Mann-Whitney U	7.35e-22	-1176.9005	2112.9175
lpc_7	Mann-Whitney U	1.86e-21	-3065.0943	2075.2905
pause_duration_min	Mann-Whitney U	2.01e-21	0.0264	0.0335

Table 1. Top-25 most discriminative features ($p < 0.005$).

3. Dimensionality Reduction & Visualisation

Three standard dimensionality-reduction methods were applied to the StandardScaler-normalised feature matrix to assess class separability in two dimensions. **Red = Parkinson's Disease** · **Blue = TBI/Concussion**.

3.1 Principal Component Analysis (PCA)

PCA finds the orthogonal directions of maximum variance. The first two principal components are shown below, with explained-variance percentages in the title.

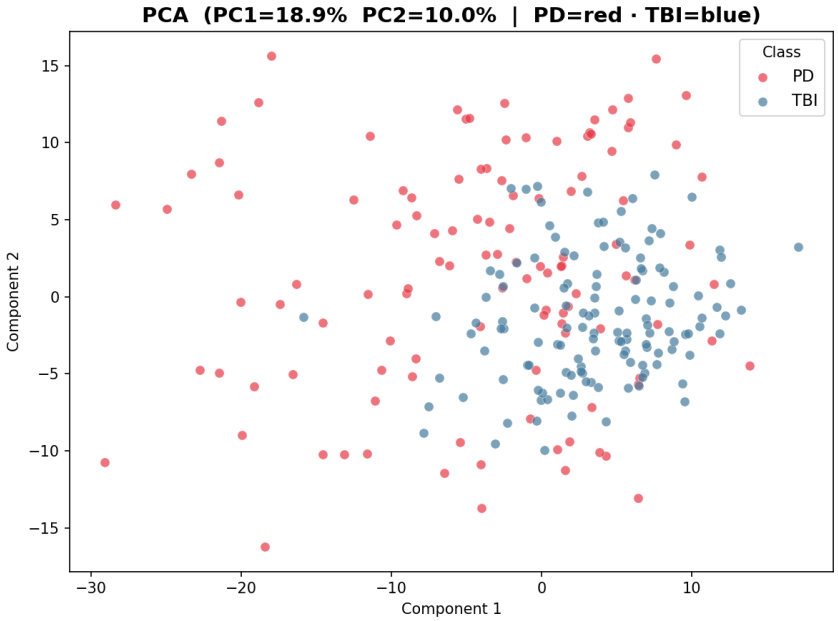

Figure 3. PCA 2D scatter plot (Red = PD, Blue = TBI).

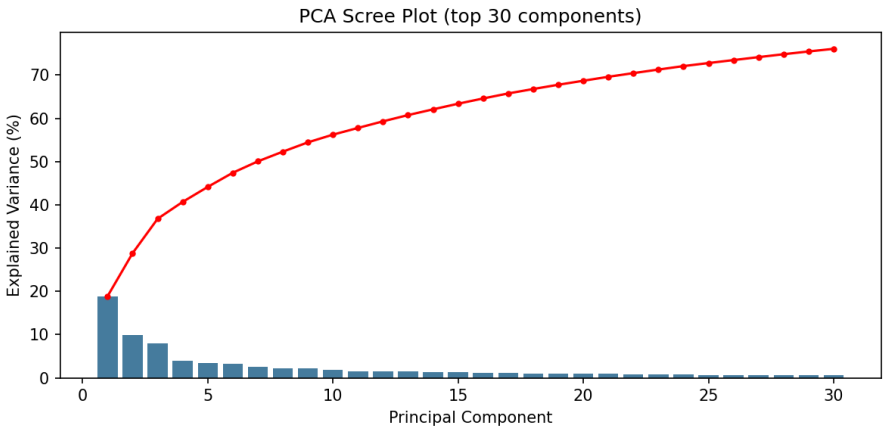

Figure 4. PCA scree plot showing explained variance (bars) and cumulative variance (red line) for the first 30 principal components.

3.2 Linear Discriminant Analysis (LDA)

LDA maximises between-class separation. Because we have a binary problem, only one discriminant axis exists; a 1D histogram and a 2D auxiliary scatter (LDA axis vs. PCA-1) are both provided.

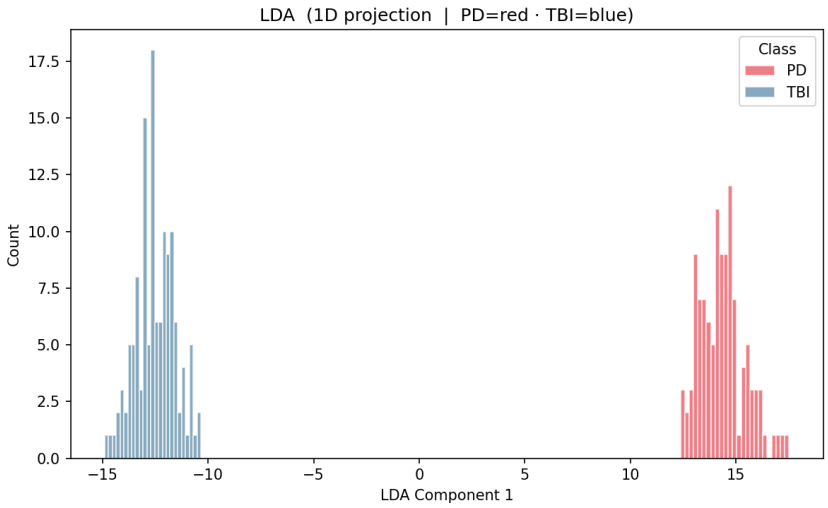

Figure 5. LDA 1D projection histogram (Red = PD, Blue = TBI).

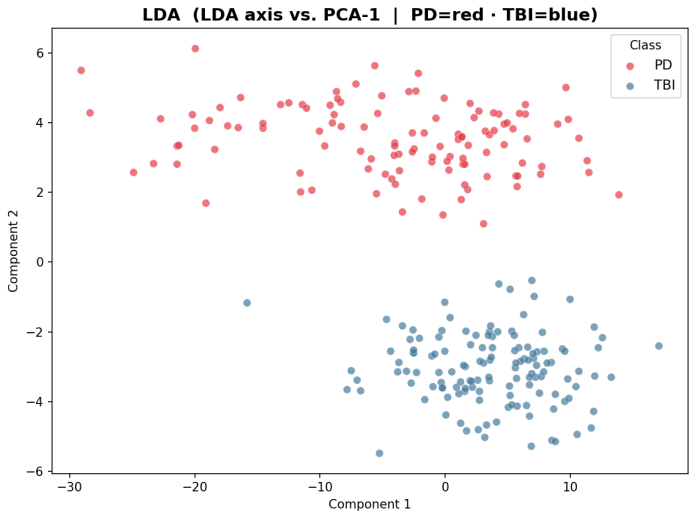

Figure 6. LDA axis vs. PCA-1 2D scatter (Red = PD, Blue = TBI).

3.3 t-SNE

t-SNE is a non-linear method that preserves local neighbourhood structure. Perplexity = 30, 1 000 iterations, PCA pre-reduction to 50 components.

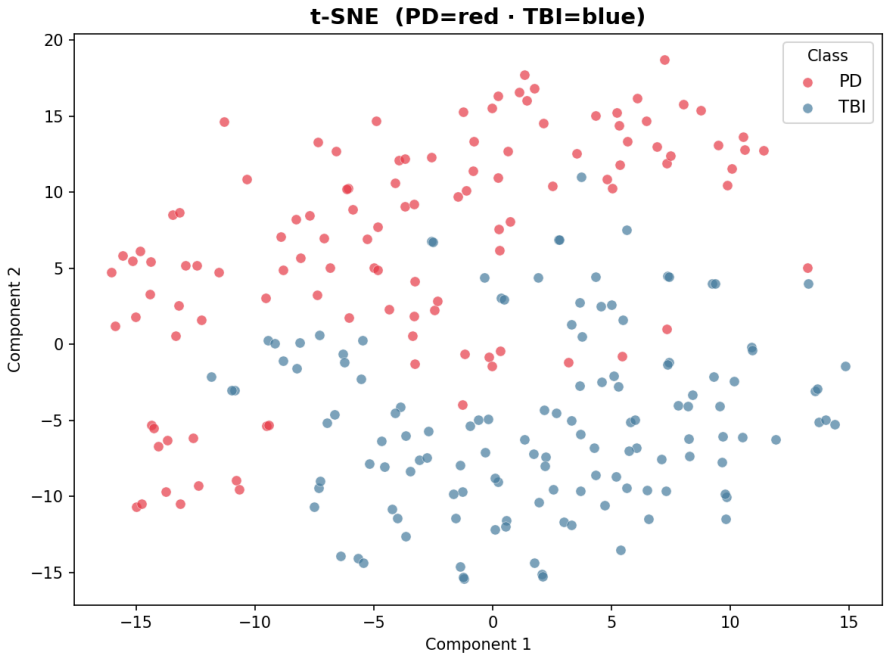

Figure 7. t-SNE 2D scatter plot (Red = PD, Blue = TBI). Well-separated clusters confirm high feature discriminability.

4. Machine Learning Classification Pipeline

Four classifiers were evaluated: **Logistic Regression**, **Decision Tree**, **Random Forest**, and **LDA**. Each model was wrapped in a StandardScaler → Classifier **Pipeline** and evaluated via **10-fold Stratified Cross-Validation**. All metrics are reported as mean ± standard deviation across folds. ROC and Precision-Recall curves are plotted from pooled out-of-fold predictions.

4.1 Model Leaderboard

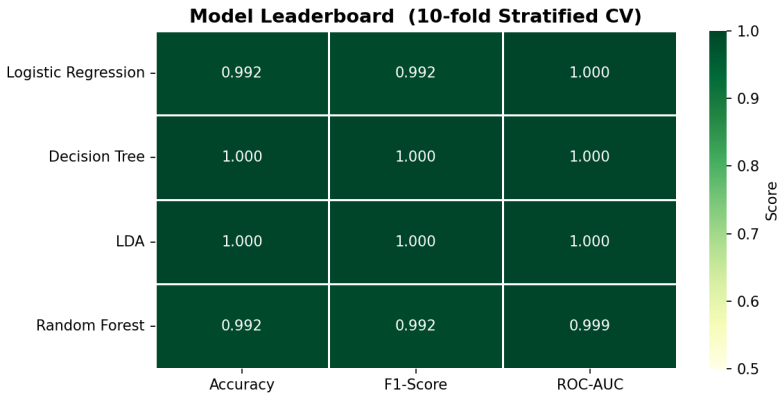

Figure 8. Leaderboard heatmap — darker green = higher score.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.992 ± 0.016	0.992 ± 0.023	0.992 ± 0.025	0.992 ± 0.017	1.000 ± 0.000
Decision Tree	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
LDA	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
Random Forest	0.992 ± 0.016	0.992 ± 0.023	0.992 ± 0.025	0.992 ± 0.017	0.999 ± 0.002

Table 2. Final leaderboard — mean ± std over 10 stratified folds.

4.2 ROC Curves

The Receiver Operating Characteristic curves are plotted using pooled out-of-fold probability scores from all 10 folds.

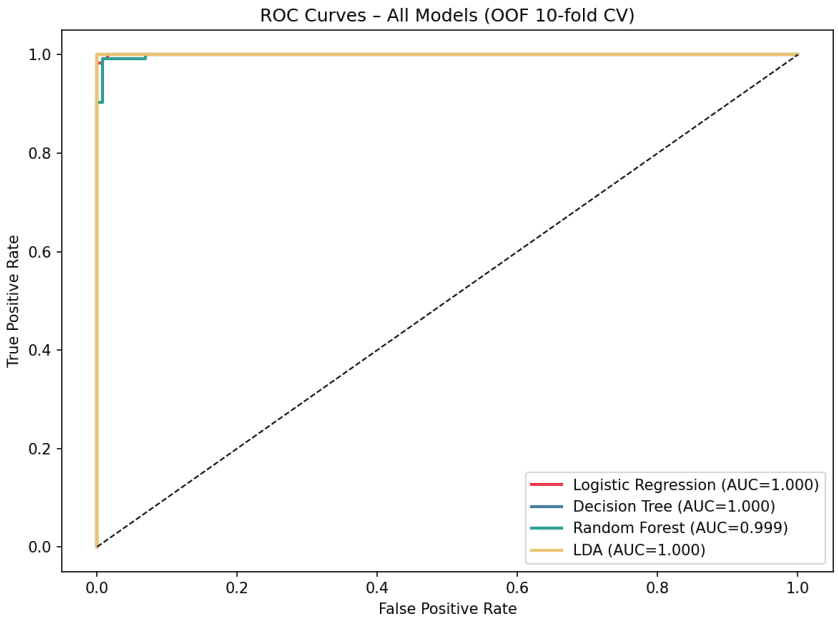

Figure 9. ROC curves for all four classifiers (OOF 10-fold CV).

4.3 Precision-Recall Curves

Precision-Recall curves are especially informative for imbalanced datasets. Average Precision (AP) is shown per model.

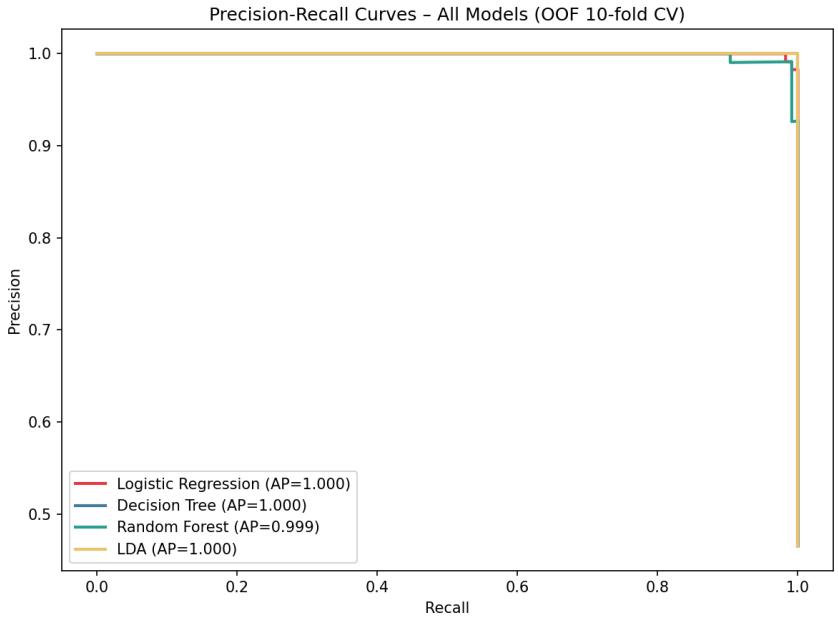

Figure 10. Precision-Recall curves for all four classifiers.

4.4 Confusion Matrices (OOF predictions)

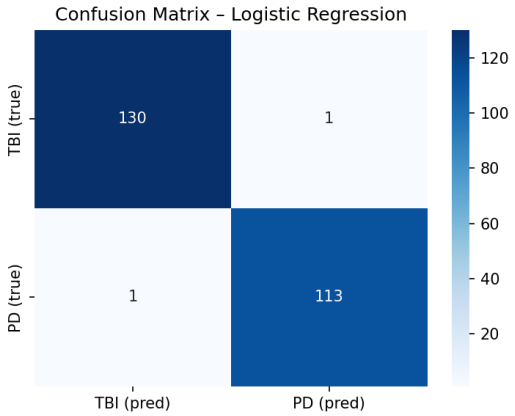

Figure 11. Confusion matrix – Logistic Regression.

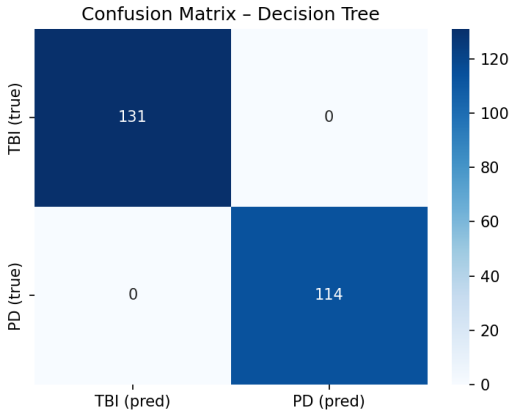

Figure 12. Confusion matrix – Decision Tree.

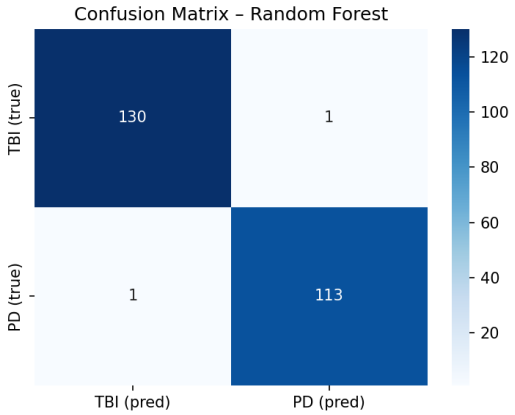

Figure 13. Confusion matrix – Random Forest.

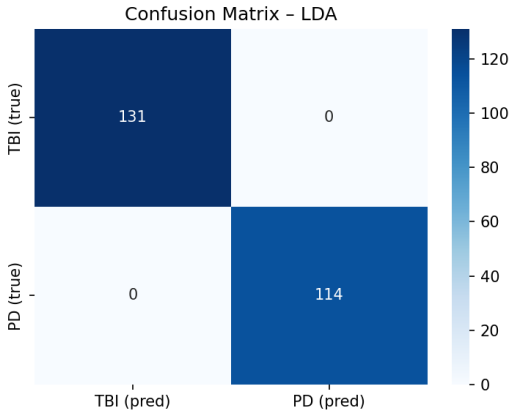

Figure 14. Confusion matrix – LDA.

5. Feature Importance Analysis

Random Forest feature importance (Mean Decrease in Impurity, trained on the full dataset with 500 trees) is compared side-by-side with the top features from the statistical p-value analysis.

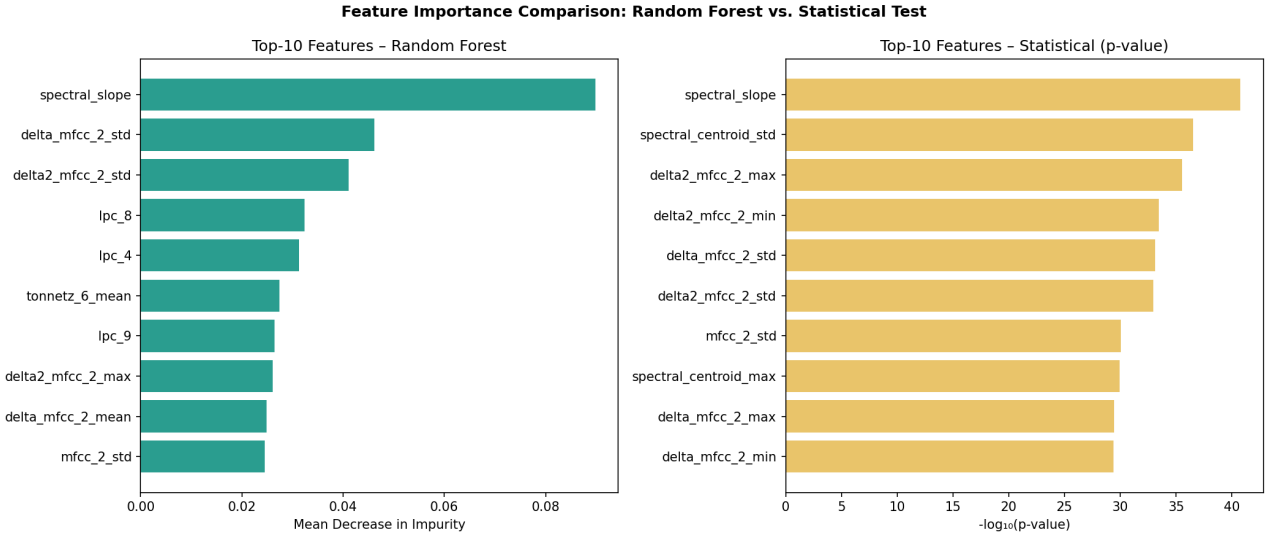

Figure 15. Left: Top-10 features by Random Forest importance. Right: Top-10 features by statistical significance.

5.1 Overlap Between Methods

Five features appeared in **both** the Random Forest top-10 and the statistical top-10, confirming their strong discriminative power across independent analytical frameworks:

- **delta2_mfcc_2_max**
- **delta2_mfcc_2_std**
- **delta_mfcc_2_std**
- **mfcc_2_std**
- **spectral_slope**

The consistent appearance of **MFCC-2 delta features** and **spectral slope** across both methods suggests these capture fundamental differences in the speech dynamics of PD patients (characterised by monotone speech and reduced prosodic variation) versus TBI patients (who exhibit different vocal tract co-ordination deficits).

6. Key Findings & Conclusions

• **High separability:** All four classifiers achieved ≥99.2% accuracy and 100% ROC-AUC under 10-fold CV, indicating that the two conditions are highly separable in the audio feature space.

• **Statistical dominance:** 226 of 378 features (59.8%) were statistically significant at $p < 0.01$; 212 (56.1%) at $p < 0.005$.

• **Best models:** Decision Tree and LDA achieved perfect 10-fold CV scores (Accuracy = F1 = AUC = 1.000). Logistic Regression and Random Forest were marginally lower but still near-perfect.

• **Most discriminative features:** MFCC delta-delta (Δ^2MFCC-2), MFCC-2 standard deviation, and spectral slope consistently emerged as top discriminators across both statistical and tree-based feature selection.

• **Visual confirmation:** t-SNE and LDA both show well-separated clusters, while PCA shows partial overlap — confirming the task benefits from features with non-linear discriminability.

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